

A taxonomy of machine-learning methods for determinant and subspace selection in SQD

importance signal exploited →

generated object →

|                                      | Cheap physics prior<br>(Epstein-Nesbet PT) | Quantum-sample<br>frequency                                        | Hamiltonian coupling<br>/ variational                   | Reward-proportional<br>(RL / flow)                               |
|--------------------------------------|--------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------|
| Sampling circuit                     |                                            |                                                                    |                                                         | GQE / Generative-QSCI<br>[Kemmoku 2026]                          |
| Determinant<br>distribution          |                                            | RBM PIGen-SQD<br>[Maitra 2025]<br>QSCI-RBM (DMET)<br>[Maitra 2026] |                                                         |                                                                  |
| Individual<br>determinants (in loop) |                                            |                                                                    | HI-NQS [2026]<br>NQS-SC [2026]<br>QiankunNet / NNQS-SCI | GFlowNet — valid- <i>N</i> by<br>construction (this review)<br>★ |
| Determinant label<br>/ ranking       | MLCI [Coe 2018]                            |                                                                    | cSQD [2026]<br>HAAR-SCI [2025]<br>RCI rank [2026]       |                                                                  |

- Generative (distribution / NQS)
- Discriminative (classifier / ranking)
- Circuit design
- Reward-proportional flow (whitespace)